

Automation of Slot Insulating Machine in Stators

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Abstract

The higher demand for automobiles in the recent years have owed many manufacturers to produce the best products in the automobile market. Starting from wheels to engines, quality and precision are the key factors that determine their superiority among the others. Similarly working of engines requires certain external supports which includes the activity of a stator. The project covers the slot insulation part of stator which is carried out using an automated slot insulation machine. The slot insulation process for stators is to provide an insulation between the copper windings and silicon steel lamination. It is important that high quality machines and tools are used in manufacturing process to achieve this condition. The slots of the stator are insulated using an insulating paper which is inserted at a higher speed. The insulating paper has to be precisely cut and inserted at a faster rate. The primary objective of our project is to integrate a slot insulation machine in the stator winding assembly line.

Keywords: PLC, Stator, Copper winding, Silicon steel, Servo drive, Capstan, MAC cards

Introduction

Since the turn of the century, the global recession has affected most businesses, including industrial automation. Because of the relatively small production volumes and huge varieties of applications, industrial automation typically utilizes new technologies developed in other markets. Automation companies tend to customize products for specific applications and requirements. So, the innovation comes from targeted applications, rather than any hot, new technology. Over the past few decades, some innovations have indeed given industrial automation new surges of growth: The programmable logic controller (PLC) was designed to replace relay-logic; it generated growth in applications where custom logic was difficult to implement and change. The PLC was a lot more reliable than relay-contacts, and much easier to program and reprogram. The PLC has had a long and productive life some three decades and has now become a commodity.

Automation software has had its day and can't go much further. No "inflection point" here. In the future, software

will embed within products and systems, with no major independent innovation on the horizon. The plethora of manufacturing software solutions and services will yield significant results, but all as part of other systems. Industrial automation can and will generate explosive growth with technology related to new inflection points. Automated factories and processes are too expensive to be rebuilt for every modification and design change so they have to be highly configurable and flexible. To successfully reconfigure an entire production line or process requires direct access to most of its control elements switches, valves, motors and drives down to a fine level of detail.

The primary objective of the project is to replace the conventional machine with up to date technology. This reflects on the performance and production rate of the machine, resulting in increased profit. The conventional slot insulation machine requires a manual labor near it all the time to pick and place the capstan. The insertion rate of the machine is 1.5 sleeves per second. The new machine can be operated at 3.5 to 5 sleeves per second. The capstan can be picked and placed in the assembly line using

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How to cite this article: Chakravarthy SR, Aaghash R, Dharshan S et al. Automation of Slot Insulating Machine in Stators. *J Adv Res Embed Sys* 2018; 5(3): 7-10.

a gripper. This will greatly reduce the cycle time and cut down the additional cost of using a manual labor. To replace manual labor and to relieve them from repeated work. The repeated work can cause fatigue which might result in minor errors paving a way for breakdown and reduced production. Also, little training is generally required, wages may not be very competitive. The work itself can also be extremely repetitive and monotonous, offering little in the way of mental stimulation and creative critical thinking.

Literature Survey

Hu Xing, Huan Jia and Liu Yanqiang [1] stated that SERCOS interface (Serial Real Time Communication Specification) is known as a universal 'motion control' interface, while EtherCAT (Ethernet for Control and Automation Technology) is a high-speed, real-time, topology-flexible industrial ethernet for data communication in process automation and factory automation control system. In order to implement the SERCOS interface on the EtherCAT network, a SERCOS over EtherCAT (SoE) model is described in this paper.

Shi Hongyu, Feng Yong and Chen Na [2] presented a distributed digital motion control system based on SERCOS (Serial Real-time Communication System). The transmission medium is optic fibre, which guarantees the system is noise immunity and high speed. The international standardize bus-SERCOS is applied to the system. The designed DSP program implements to control the motor and communicate with host computer.

Robert W. Wieseman [3] describes that Insulation resistance of the windings of an electric machine may be used as a barometer for indicating the condition of the windings. On the basis of tests on many machines of different types it is recommended that all insulation resistance readings be taken after the application of 500 volts (d-c) for one minute. The slope of the resistance-time curve at that point gives a combined indication of the moisture content and the quality of the insulation.

B.K. Gupta, G.C. Stone and J. Stein [4] results that over-potential or hipot tests are frequently applied to machine stator windings to assure that the electrical insulation in windings is fit for service.

Slot Insulation Machine in Stator Line

The slot insulation machine was pre-existing in the stator assembly line. The function of the machine is to pick the capstan from the pallet and to insert insulation sheets in the slots; The finished capstan will be placed back in the pallet. The slot insulation machine inserts the insulation paper in certain mechanism. The insertion mechanism will be explained below. The existing machine and its control panel is shown below. The slot insulation machine is a high-speed operating machine which inserts the insulating

paper at a rate of 3 sleeves per second. It performs this job with the help of six sequential operations of six servo motors each assigned to perform certain tasks. The control system of these motors involves the likes of drives, PLC, circuit breaking systems and the power supply.

The machine traces back its origin before 30 years which includes servo motors with separate encoders and a control panel with very old servo drives and Mac cards to process the encoder values. The B&R NT 43 PLC along with Axis control cards was used for controlling the Bosch servo drives. The digital inputs and outputs were controlled using Bosch CL300. The usage of MAC cards for processing encoder values led to a series of connections which was very time consuming and any hardware issues could not be met immediately due to the lack of service facilities. The PLCs and Drives were obsolete and are out of spares and service. This resulted in breakdown of machine often. The machine was able to achieve a cycle time of 12 seconds.

Components Selection

Selection of PLC

There were a number of suggestions in PLC and a few of them were studied which could suit for the slot insulation machine. The Rexroth PLCs manufactured by Bosch were more suitable for the application due to its robustness and integrability. The PC-based IndraLogic V PLC systems are consistently available with standard run-time system on different designs and performance classes. Whether you are using panel PCs or separate PC and control units – IndraLogic V provides all kinds of freedoms for controlling and visualizing the applications. The tiered device portfolio with extremely robust, industrial-strength design in combination with the real time operating systems allows reliable use in a wide variety of applications. The PLC series IndraLogic L is based on the scalable control platform CML and is available in different performance classes in a uniform and compatible system design. The compact design with terminal technology and easy mounting on the DIN rail can be used in any automation environment. The IndraControl XM control platform is available for a wide variety of different motion logic applications. Local connection of the modules from the new IndraControl S20 I/O family expands the control system flexibly to a high-performance process link. As standard, decentralized I/Os, drives, and other system peripherals are connected via Sercos. Optional extension modules open up further integration options. The service-friendly design is an easy way to reduce installation and commissioning costs.

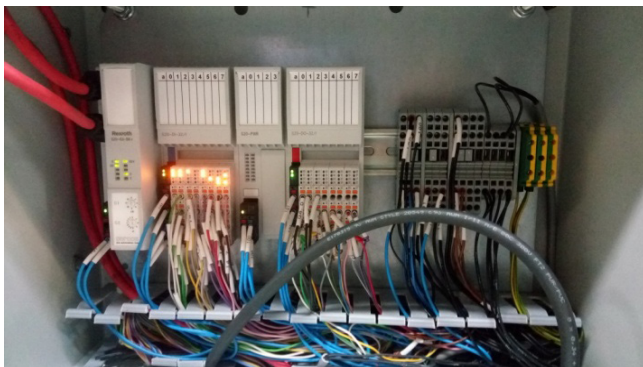
Selection of Remote I/O Module

The slot insulation machine and the control panel are in the two different places. If all I/O wires are taken to the control panel, there will be lot of wires used and it would

be difficult to change the sensors in case of any failure in the sensor or the break in the wire. So, a S20 remote I/O module is installed at the place of the machine, to which the I/O are connected and the module communicates with PLC through Sercos communication which is the fastest and one-wire communication.

Rexroth S20 I/O module, Bosch

The I/O module discussed here is Rexroth S20 module which is regarded as highly protective module and it has proved to be the highly secure and fast way of communication in the modern industry. These modules are designed for use within an IndraControl S20 station. The digital output modules are used to output 24 V DC signals. Actuators can be connected in 1-wire technology.



Servo Motor

The machine has the six axes. All the six axes are running synchronizing at the same time with the different rate of speed and load. In order to meet this requirement, the Rexroth IndraDyn S Servo motors are used for the actuation of machine. All motors are controlled by the Servo drives. The servo drive is controlled by the PLC. Servo implies an error sensing feedback control which is utilized to correct the performance of a system. It also requires a generally sophisticated controller, often a dedicated module designed particularly for use with servomotors. Servo motors are DC motors that allow for precise control of angular position. They are actually DC motors whose speed is slowly lowered by the gears. The servo motors usually have a revolution cutoff from 90° to 180°. A few servo motors also have a revolution cutoff of 360° or more. But servo motors do not rotate constantly.



Rexroth IndraDyn S Servo Motor
Indra Control CS Servo Drive

A servo drive is a special electronic amplifier used to power electric servomechanisms. A servo drive monitors the feedback signal from the servomechanism and continually adjusts for deviation from expected behavior. A servo drive receives a command signal from a control system, amplifies the signal, and transmits electric current to a servo motor in order to produce motion proportional to the command signal.



IndraControl Drive CS
Sercos Communication

Sercos has been one of the leading bus systems for industrial applications for more than 25 years. The real-time technology with its millionfold proven quality, universal application possibilities and high security of investments make the Sercos automation bus the first choice in mechanical engineering and construction. Industrial automation requires real-time-enabled and manufacturer-independent communication solutions. Different kinds of automation devices must be able to be easily and universally connected. The open, IEC standard universal bus for Ethernet-based real-time communication,

Sercos III, meets these demands today with a wide variety of benefits. In operation, Sercos communication takes place cyclically as master/slave communication at the cycle time selected at initialization.

Programming and Analysis

IndraWorks allows you to tackle all PLC-based automation and drive commissioning tasks in a uniform and intuitively operable software environment. All of the basic tools required, including the homogeneously integrated CoDeSys V3, are available on a continuous basis. As a user, you benefit from quick and transparent access to all the functions and system data of the automation components. IndraWorks offers continuous operation based on current Windows technologies and extensive wizards for the project planning of controls, drives, and peripherals.

Production loss can be minimized with "Alarm Analysis" function. When an error occurs, an operator can easily seek and check the condition on-site, just by touching the alarm message to call up various error-related data in chronological order. An alarm analysis screen with timing chart of alarm-related device addresses can be simply configured only through a few steps on GP-Pro EX software for easy troubleshooting, debugging and/or design changes of production equipment to reduce downtime for enhanced productivity. Operation Log - Recording the operator's operation. Operations are saved to a log, so you can see who did what, and when. The log can also be used for a trend analysis of erroneous operation of operators. Change display sizes easily with Resolution Converter. GP-Pro EX's Resolution Converter automatically converts project files into different resolutions. This significantly reduces man-hours associated with changing display sizes.

Programming function relieves PLC memory burden. GP-Pro EX supports logic programs, which let you create programs in a familiar ladder program format, and D-Script, a proprietary scripting language that offers similar functionality to sophisticated programming languages. This lowers the burden placed on the PLC and enables high-level screen design. Logic Program - Program with a familiar ladder language. Screens and logic programming can be edited with the same software, so editing between parts and logic elements can be performed via drag and drop. Controller addresses can be directly written, allowing you to reduce development time.

Result and Conclusion

The Servo Drives and the I/O can be controlled using the same PLC. The Encoders are inbuilt in the Servo Motors itself. The Communication is upgraded to suffice faster transfer rate up to 100Mbps. The Overall Equipment Effectiveness (OEE) is increased and reduced technical loss. Low cost in-house execution and Improved Product quality. Online problem diagnosis is possible. External Service support and spares are available.

Having been exposed to greater industrial technologies and learning what the industry expects from young talents, the project was a great experience and a pleasure way of learning. The study of working of assembly line gave a clear idea on how assembly should be done to achieve higher production in lesser time. The slot insulation project gave in depth knowledge about new PLC and the advancements in drives recently. The upgrade of existing machine is a quite good example of cost efficiency without compromising the performance. The vast variety of motors and different assembly lines have nurtured the habit of thinking about new ideas and what can be done for the betterment of the existing assembly line. The overall experience in the industry have benefited in such a way that it will stay as an unforgettable lesson in the future irrespective of the field of work.

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Date of Submission: 2018-06-16

Date of Acceptance: 2018-07-05